

Cambridge International AS & A Level

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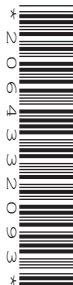
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FURTHER MATHEMATICS

9231/31

Paper 3 Further Mechanics

October/November 2022

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

[3]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

- 2 A light elastic string has natural length a and modulus of elasticity $4mg$. One end of the string is fixed to a point O on a smooth horizontal surface. A particle P of mass m is attached to the other end of the string. The particle P is projected along the surface in the direction OP . When the length of the string is $\frac{5}{4}a$, the speed of P is v . When the length of the string is $\frac{3}{2}a$, the speed of P is $\frac{1}{2}v$.

(a) Find an expression for v in terms of a and g . [4]

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(b) Find, in terms of g , the acceleration of P when the stretched length of the string is $\frac{3}{2}a$. [2]

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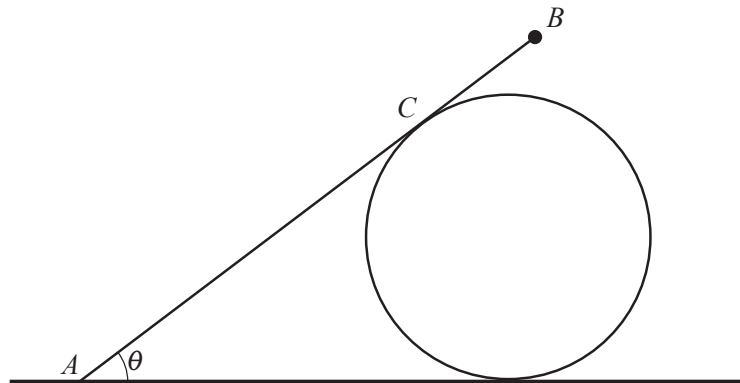
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A particle of weight kW is attached to the rod at B . The rod is about to slip. The normal reaction between the rod and the cylinder is N .

- (a) Show that $N = \frac{8}{15} W(1 + 2k)$. [2]

This image shows a full page of a handwriting practice worksheet. It consists of multiple sets of three horizontal dashed lines, providing a guide for letter height and placement. The lines are evenly spaced across the entire page, leaving ample room for writing practice. There is no text or other markings on the page.

[Turn over

- 4 A particle of mass 0.5 kg moves along a horizontal straight line. Its velocity is $v\text{ ms}^{-1}$ at time $t\text{ s}$. The forces acting on the particle are a driving force of magnitude 50 N and a resistance of magnitude $2v^2\text{ N}$. The initial velocity of the particle is 3 ms^{-1} .

(a) Find an expression for v in terms of t . [7]

[illegible]

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(b) Deduce the limiting value of v . [1]

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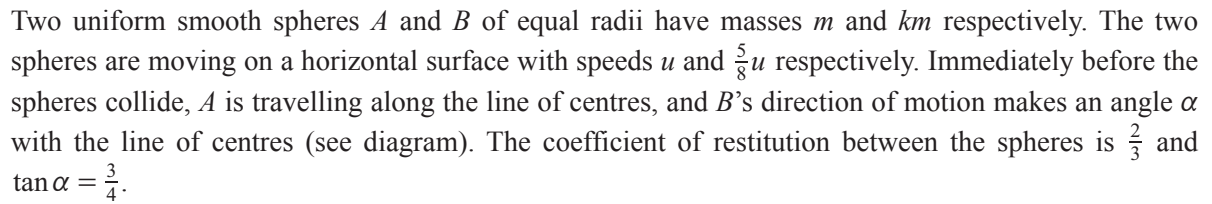
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- [illegible]

At Q the particle P becomes detached from the string.

- (b) In the subsequent motion, find the greatest height reached by P above the level of the lowest point of the circle. [4]

[illegible]



(a) Find the value of k . [4]

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- [illegible]

- 7 A particle P is projected with speed $V \text{ m s}^{-1}$ at an angle 75° above the horizontal from a point O on a horizontal plane. It then moves freely under gravity.

(a) Show that the total time of flight, in seconds, is $\frac{2V}{g} \sin 75^\circ$. [2]

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A smooth vertical barrier is now inserted with its lower end on the plane at a distance 15 m from O . The particle is projected as before but now strikes the barrier, rebounds and returns to O . The coefficient of restitution between the barrier and the particle is $\frac{3}{5}$.

(b) Explain why the total time of flight is unchanged. [1]

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- (c) Find an expression for V in terms of g . [7]

[illegible]

Additional page

If you use the following page to complete the answer to any question, the question number must be clearly shown.

[illegible]

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